Date Day

Class A → 0.374
Class B → 0.436 ✓ Higher probability

"DATA FUSION TECHNIQUE"

→ It is the process of combining data from multiple sources to produce more accurate, reliable & complete info rather than using a single source

→ TYPES OF DATA FUSION

(1) Early Fusion (Feature-level Fusion)
↳ Raw data or extracted features from multiple sources are combined at the begining before training.
↳ The model is trained on the fused feature vector
↳ Advantages
- Uses full original data
- Can give very accurate results

↳ Process :
(1) Features extracted from each modality
(2) After extraction they are concatenated into a single

Disadvantages:
- Combined features becm very large
- diff data may have diff scales, timing

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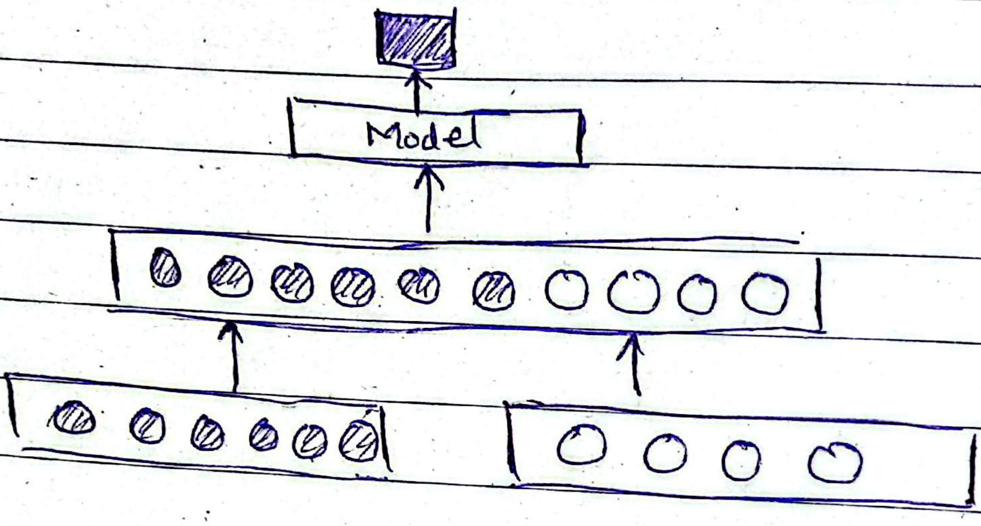
feature vector

③ Concatenated feature vector is used to train



② Late Fusion (Decision Level Fusion)

↳ Each data source is processed independently and final decision/predictions combined afterward

↳ For eg: In a video classification task,

m1 → trained on video frames

m2 → trained on audio

m3 → trained on textual description

- Once all are trained, they generate their predicti
- Decisions of each model is then aggregate using techniques such as voting, averaging, or weighted summation

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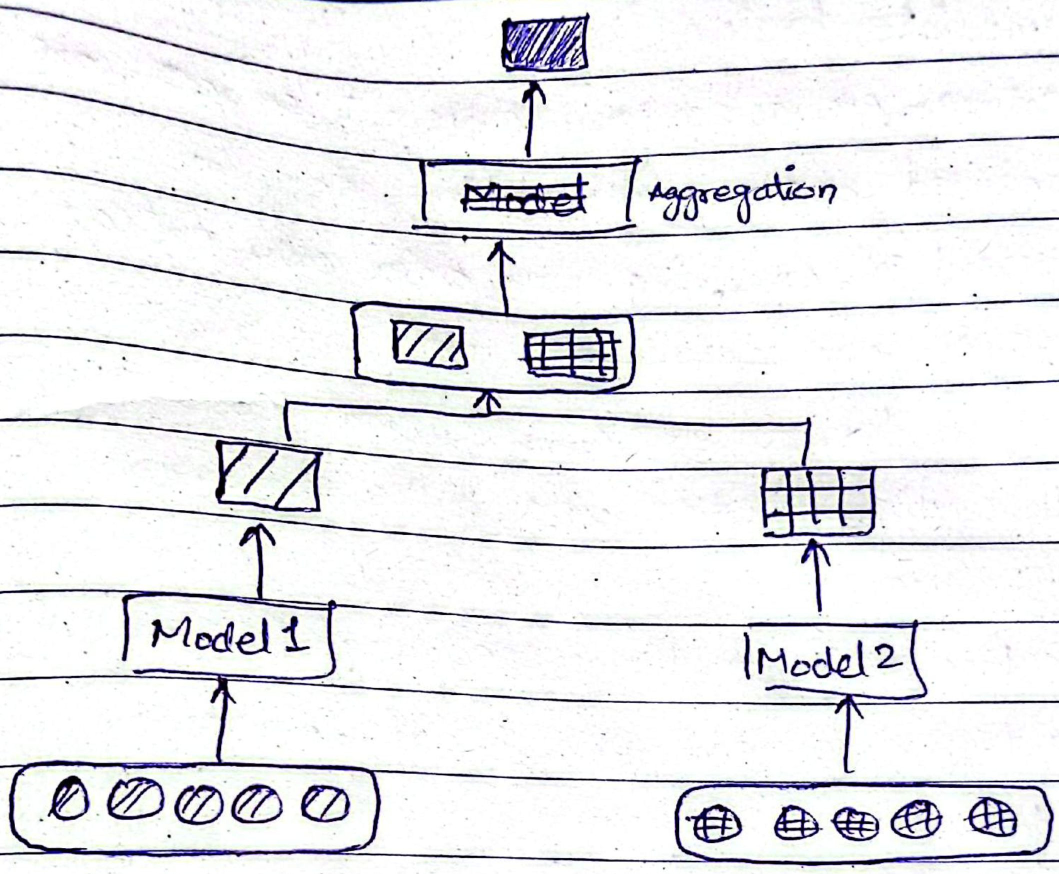


↳ Advantages:

↳ Independent Models easier to design

↳ One source missing doesnot break sys

↳ Models can be updated independently

↳ High Dimensional feature space issue resolved

↳ Disadvantages

↳ Sometimes lower accuracy as compared to early f—

↳ interaction b/w modalities not learned properly

③ Intermediate Fusion

↳ Features extracted from different sources are combined before making final decision

↳ Instead of combining raw data only extracted features are ~~consumed~~ fused

↳ reduces computational costs

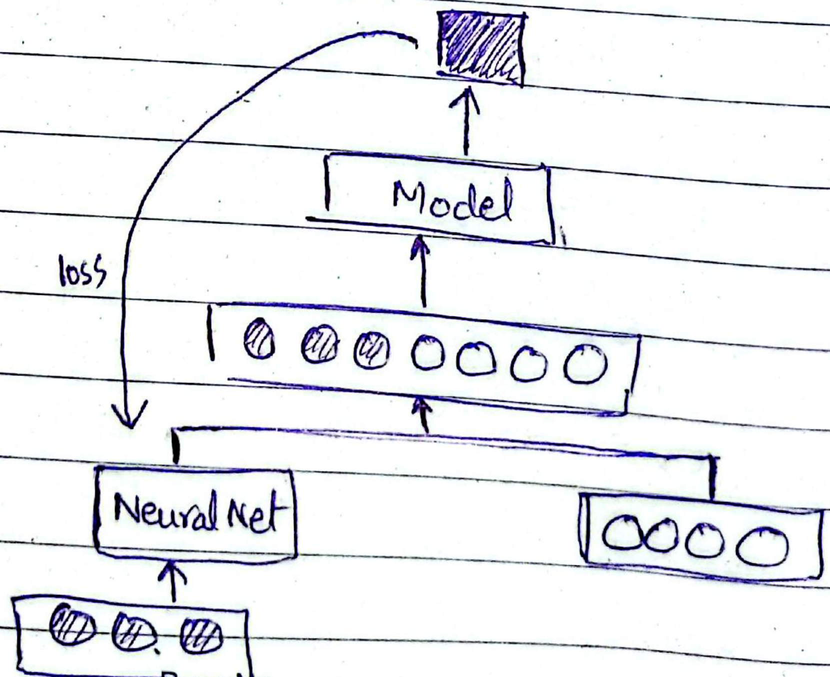
↳ Allows diff types of data to be combined

↳ Advantages

- Each modality learns specialized features list
- Diff models can process diff data types

↳ Disadv

- Requires sophisticated architecture
- Multiple networks involved
- ~~Fusin~~



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④ Attention Based Fusion
↳ · Advance fusion technique where model learns
  ↳ which input/source is more important
  ↳ where to focus
  ↳ How much weight each modality should recv
↳ Doesnot treats each data source equally
↳ Helps focus on important features

→ MODALITY CHARACTERISTICS
① Representation
  Images → 2D Array
  Text → words token
  Audio → waveforms, frequency signals
② Dimensionality
  Images → high-dim (thousands of pixels)
  Text → variable len
  Audio → time-series signals
③ Temporal Nature (whether data changes over time)
  Static → images
  Dynamic → audio, video, sensor data

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(4) Correlation

- Lip movement ⟷ speech
- Object image ⟷ label (text)